Reasoning test

For admission to Class 10

About the test

- 15 questions in 15 minutes: 5 each of maths, English and science, mixed together.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers

The questions test reasoning, not memory, using ideas taught up to Class 9. Chapters are from the NCERT (CBSE) books a student joining Class 10 in 2027 will have used; other boards teach the same ideas under their own names.

Maths

Class 6 · Mathematics

- *Whole Numbers* – number patterns and series
- *Playing with Numbers* – primes, factors, HCF and LCM
- *Fractions*
- *Mensuration* – perimeter and area
- *Ratio and Proportion*

Class 7 · Mathematics

- *Fractions and Decimals* – fractions
- *Data Handling* – averages (mean)
- *Simple Equations*
- *Comparing Quantities* – percentages, profit and loss, discounts, interest; ratio and proportion
- *Perimeter and Area*

Class 7 · Science

- *Motion and Time* – speed, distance and time

Class 8 · Ganita Prakash

- *A Square and A Cube* – squares and cubes
- *Proportional Reasoning-1* – ratio and proportion
- *Fractions in Disguise* – percentages, profit and loss, discounts, interest
- *Proportional Reasoning-2* – ratio and proportion
- *Tales by Dots and Lines* – averages (mean)

Class 9 · Ganita Manjari

- *Measuring Space: Perimeter and Area* – perimeter and area
- *The Mathematics of Maybe: Introduction to Probability* – probability
- *Predicting What Comes Next?: Exploring Sequences and Progressions* – number patterns and series

Reasoning, not a textbook chapter: Clocks and calendars · Number codes and puzzles · Everyday units · Multi-step word problems · Time and work, pipes and tanks · Problems on ages · Mixtures · Overlapping groups (Venn diagrams)

English

Grammar, vocabulary and reasoning; nothing from the English reader: Vocabulary: meaning in context, opposites, idioms · Grammar: agreement, prepositions, tenses, linking words · Completing and ordering sentences · Analogies, odd one out, letter series and codes · Reading: inference, tone, assumptions, judging an argument · Logical deduction: statements, arrangements, family relations

Science

Class 6 · Science

- *Fun with Magnets* – magnets
- *Light, Shadows and Reflections* – light, shadows and plane mirrors
- *Electricity and Circuits* – electric circuits

Class 7 · Science

- *Heat* – heat and conductors
- *Acids, Bases and Salts* – acids, bases and indicators
- *Physical and Chemical Changes* – physical and chemical changes, rusting
- *Nutrition in Plants* – photosynthesis and plant respiration
- *Respiration in Organisms* – respiration; photosynthesis and plant respiration
- *Motion and Time* – speed
- *Light* – light, shadows and plane mirrors

Class 8 · Curiosity

- *Exploring Forces* – floating, sinking and density; mass and weight
- *Pressure, Winds, Storms, and Cyclones* – force and pressure, pressure in liquids
- *Particulate Nature of Matter* – evaporation and condensation; change of state
- *Nature of Matter: Elements, Compounds, and Mixtures* – elements, compounds, mixtures and solutions
- *The Amazing World of Solutes, Solvents, and Solutions* – elements, compounds, mixtures and solutions
- *How Nature Works in Harmony* – food chains

Class 9 · Exploration

- *How Forces Affect Motion* – inertia, action and reaction
- *Work, Energy and Simple Machines* – energy changes
- *Exploring Mixtures and their Separation* – elements, compounds, mixtures and solutions
- *Sound Waves – Characteristics and Applications* – sound, and the speeds of light and sound

Reasoning, not a textbook chapter: Reading an experiment and fair tests · Applying a principle to a new situation

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS

   What is the angle between the hands of a clock at exactly 3 o'clock?

   A 45°
   B 60°
   C 75°
   D 90°

2. ENGLISH

   The general's plan was simple but ingenious. As used here, "ingenious" most nearly means

   A dangerous
   B clever
   C costly
   D careless

3. SCIENCE

   As a ball falls freely towards the ground, which energy change takes place?

   A potential energy to kinetic energy
   B kinetic energy to potential energy
   C kinetic energy to chemical energy
   D heat energy to potential energy

4. MATHS

   The price of a bag rises from ₹250 to ₹300. What is the percentage increase?

   A 15%
   B 20%
   C 25%
   D 50%

5. ENGLISH

   Choose the correct words: The bundle of letters ___ on the table.

   A were lying
   B are lying
   C was lying
   D have lain

6. SCIENCE

   An astronaut has a mass of 60 kg on Earth. What is her mass on the Moon?

   A 0 kg
   B 10 kg
   C 60 kg
   D 360 kg

7. MATHS

   A car uses 8 litres of petrol to travel 96 km. How many litres will it need to travel 156 km?

   Type-in answer: _____

8. ENGLISH

   Ear is to hearing as eye is to ___

   A tears
   B face
   C colour
   D sight

9. SCIENCE

   An ant bite injects an acid into the skin. Which of these would best relieve the sting?

   A a few drops of vinegar
   B baking soda paste
   C fresh lemon juice
   D plain salt water

10. MATHS

₹5,000 is deposited at 10% interest per year, compounded yearly. What will the amount be after 2 years?

A ₹5,500
B ₹6,000
C ₹6,050
D ₹6,500

11. ENGLISH

"The bus was late, the rain soaked us and the shop was shut. What a perfect day!" The speaker's tone is

A sarcastic
B sincere
C hopeful
D indifferent

12. SCIENCE

40 g of sugar is dissolved in water to make 500 ml of solution. What is the concentration, in grams per litre?

Type-in answer: _____

13. MATHS

In a class of 50 students, 30 play cricket, 25 play football and 8 play neither game. How many play both?

A 8
B 13
C 17
D 22

14. ENGLISH

"New traffic lights were installed in May and road accidents fell in June, so the lights prevent accidents." Which fact, if true, most weakens this argument?

A The lights cost the town a lot
B Some drivers dislike waiting at lights
C Schools closed in June, cutting traffic
D Accidents also happen late at night

15. SCIENCE

A burn from steam at 100 °C is usually worse than one from boiling water at 100 °C. This is because steam

A is at a higher temperature than water
B moves faster and hits the skin harder
C carries germs that the water does not
D gives out extra heat as it condenses

Answers

1	**D. 90°**	The clock face is 360°, and each hour mark is 30° apart. From 12 to 3 is three hour marks, so the angle is 3 × 30° = 90°.
2	**B. clever**	Ingenious means clever and inventive. "Simple but" signals a quality that is praised, which rules out careless and dangerous.
3	**A. potential energy to kinetic energy**	As the ball falls it loses height, so it loses potential energy, and it speeds up, so it gains kinetic energy.
4	**B. 20%**	The rise is ₹50, and the percentage is worked out on the original price: 50 ÷ 250 = 0.2, which is 20%.
5	**C. was lying**	The subject is "the bundle", which is singular; "of letters" only describes it. So the verb is "was lying".
6	**C. 60 kg**	Mass is the amount of matter and does not change from place to place. Her weight on the Moon is about one sixth, but her mass stays 60 kg.
7	**13**	The car runs 96 ÷ 8 = 12 km on each litre, so 156 km needs 156 ÷ 12 = 13 litres. This is a type-in question.
8	**D. sight**	The pair links an organ to the sense it provides. The ear gives hearing and the eye gives sight.
9	**B. baking soda paste**	An acid is neutralised by a base. Baking soda is a mild base; vinegar and lemon juice are acids and would not help.
10	**C. ₹6,050**	After year 1: 5,000 + 500 = ₹5,500. Year 2 earns 10% of 5,500, which is ₹550, giving ₹6,050. ₹6,000 is simple interest.
11	**A. sarcastic**	Everything described went wrong, so calling it a perfect day means the opposite of what is said. That is sarcasm.
12	**80**	500 ml is half a litre. 40 g in half a litre is the same as 80 g in a full litre. This is a type-in question.
13	**B. 13**	50 − 8 = 42 students play at least one game. 30 + 25 = 55 counts the players of both games twice, so 55 − 42 = 13 play both.
14	**C. Schools closed in June, cutting traffic**	If traffic itself fell in June, that alone could explain fewer accidents, so the lights may not be the cause. The others do not touch the cause.
15	**D. gives out extra heat as it condenses**	Both are at 100 °C, but steam releases a large amount of extra heat (latent heat) when it turns back into water on the skin.